

KITABU QUEST

P4

MATHEMATICS



Cover scene

learners using counters and measuring tools with a friendly mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

4

Title Page

KITABU QUEST P4 Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps P4 learners study Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use objects, drawings, number sentences, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Mathematical operations on whole numbers up to 100,000
2. Positive and negative integers
3. Classifying numbers by their properties
4. Fractions with the same denominator
5. Decimal numbers
6. Length measurements
7. Capacity Measurements
8. Mass measurements
9. Area and land measurements
10. Time
11. Number patterns
12. Solving problems involving missing number in addition and subtraction

As you finish each topic, return to the success criteria and correct one answer before moving on.

Mathematical operations on whole numbers up to 100,000: Lesson Opener

Learning focus: Mathematical operations on whole numbers up to 100,000

Country link: a Rwandan school, market, farming, building, transport, or data problem for Mathematical operations on whole numbers up to 100,000.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- read, write, compare and make calculations on whole numbers up to 100,000
- name place values up to 100,000 in both spoken and written form in English
- identify the place values of numerals
- read written numerals correctly in English
- explain the process of addition of Two numbers of 5 digits with or without carrying
- apply place value to compare numbers of any size

Inquiry questions:

- How can mathematical operations on whole numbers up to 100,000 help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Mathematical operations on whole numbers up to 100,000: Learn

Mathematical operations on whole numbers up to 100,000 teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Mathematical operations on whole numbers up to 100,000
- Mathematical
- operations
- whole
- numbers
- 100,000
- method
- model

Mathematical operations on whole numbers up to 100,000:

Worked Example

Worked example for Mathematical operations on whole numbers up to 100,000: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Mathematical operations on whole numbers up to 100,000:

Vocabulary And Study Skill

Use these words and study moves before you practise Mathematical operations on whole numbers up to 100,000. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Mathematical operations on whole numbers up to 100,000
- Mathematical
- operations
- whole
- numbers
- 100,000
- method
- model

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Mathematical operations on whole numbers up to 100,000 without a clear example, method, or evidence.

Mathematical operations on whole numbers up to 100,000: Guided Activity

Guided activity for Mathematical operations on whole numbers up to 100,000:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Mathematical operations on whole numbers up to 100,000: Practice And Correction

Practice for Mathematical operations on whole numbers up to 100,000:

Main outcome: read, write, compare and make calculations on whole numbers up to 100,000.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to mathematical operations on whole numbers up to 100,000 and solve it.

Mathematical operations on whole numbers up to 100,000: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Mathematical operations on whole numbers up to 100,000.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Mathematical operations on whole numbers up to 100,000: Outcome Check

Outcome check for Mathematical operations on whole numbers up to 100,000: Read, write, compare and make calculations on whole numbers up to 100,000.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Mathematical operations on whole numbers up to 100,000: Outcome Check

Outcome check for Mathematical operations on whole numbers up to 100,000: Name place values up to 100,000 in both spoken and written form in English.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Positive and negative integers: Lesson Opener

Learning focus: Positive and negative integers

Country link: a Rwandan school, market, farming, building, transport, or data problem for Positive and negative integers.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problems related to comparing, ordering, and finding the distance between negative and positive integers
- explain that on a number line, the number located to the right is always greater than the number to the left
- locate positive and negative numbers on the number line
- identify and compare two numbers using their positions on a number line
- calculate the distance between integers
- appreciate the importance of positive and negative numbers in real-life situations and develop confidence in locating them on a number line

Inquiry questions:

- How can positive and negative integers help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Positive and negative integers: Learn

Positive and negative integers teaches numbers below, at, and above zero.

Core idea:

- Positive numbers are greater than zero.
- Negative numbers are less than zero.
- Zero is the turning point.
- A number line helps you compare and order integers.

Rwanda example: temperature changes, money owed, points gained or lost in a game, and height above or below a reference point can use integers.

Key words:

- Positive and negative integers
- Positive
- negative
- integers
- method
- model
- unit
- answer

Positive and negative integers: Worked Example

Worked example for Positive and negative integers: The morning temperature is 3 degrees above zero. At night it drops by 7 degrees. What is the night temperature?

1. Start at +3 on a number line.
2. Move 7 steps left because the temperature drops.
3. $+3 - 7 = -4$.

Answer: 4 degrees below zero.

Positive and negative integers: Vocabulary And Study Skill

Use these words and study moves before you practise Positive and negative integers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Positive and negative integers
- Positive
- negative
- integers
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Positive and negative integers without a clear example, method, or evidence.

Positive and negative integers: Guided Activity

Guided activity for Positive and negative integers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Positive and negative integers: Practice And Correction

Practice for Positive and negative integers:

Main outcome: solve problems related to comparing, ordering, and finding the distance between negative and positive integers.

1. Solve one positive and negative integers question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to positive and negative integers and solve it.

Positive and negative integers: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Positive and negative integers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Positive and negative integers: Outcome Check

Outcome check for Positive and negative integers: Solve problems related to comparing, ordering, and finding the distance between negative and positive integers.

1. Place -3, 0, and +4 on a number line.
2. Which number is greatest?
3. Find the distance from -3 to +4.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Positive and negative integers: Outcome Check

Outcome check for Positive and negative integers: Explain that on a number line, the number located to the right is always greater than the number to the left.

1. Place -3, 0, and +4 on a number line.
2. Which number is greatest?
3. Find the distance from -3 to +4.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Classifying numbers by their properties: Lesson Opener

Learning focus: Classifying numbers by their properties

Country link: a Rwandan school, market, farming, building, transport, or data problem for Classifying numbers by their properties.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- classify numbers flexibly, seeing them as belonging to various sets
- explain the meaning of odd, even, square numbers, square root, prime numbers, multiples of numbers, factors of numbers
- list the types of numbers in each number family and explain the defining properties of each family
- solve and explain problems about classifying numbers by their properties
- classify numbers as odd, even, square numbers, prime numbers, multiples of a given number, factors of a given number
- calculate the square of a number and find the square root of a square number ≤ 100

Inquiry questions:

- How can classifying numbers by their properties help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Classifying numbers by their properties: Learn

Classifying numbers by their properties helps you break numbers into useful groups.

Core idea:

- A factor divides a number exactly.
- A multiple is made by multiplying a number by 1, 2, 3, and so on.
- Prime numbers have exactly two factors: 1 and the number itself.
- Classification is easier when you test divisibility carefully.

Key words:

- Classifying numbers by their properties
- Classifying
- numbers
- their
- properties
- method
- model
- unit

Classifying numbers by their properties: Worked Example

Worked example for Classifying numbers by their properties: List all factors of 24.

1. Test factor pairs: 1×24 , 2×12 , 3×8 , 4×6 .
2. No whole-number factor pair is missing between 4 and 6.
3. Write the factors in order.

Answer: 1, 2, 3, 4, 6, 8, 12, 24.

Classifying numbers by their properties: Vocabulary And Study Skill

Use these words and study moves before you practise Classifying numbers by their properties. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Classifying numbers by their properties
- Classifying
- numbers
- their
- properties
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Classifying numbers by their properties without a clear example, method, or evidence.

Classifying numbers by their properties: Guided Activity

Guided activity for Classifying numbers by their properties:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Classifying numbers by their properties: Practice And Correction

Practice for Classifying numbers by their properties:

Main outcome: classify numbers flexibly, seeing them as belonging to various sets.

1. Solve one classifying numbers by their properties question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to classifying numbers by their properties and solve it.

Classifying numbers by their properties: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Classifying numbers by their properties.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Classifying numbers by their properties: Outcome Check

Outcome check for Classifying numbers by their properties: Classify numbers flexibly, seeing them as belonging to various sets.

1. List all factors of 18.
2. List the first five multiples of 6.
3. Circle any prime numbers in 2, 3, 4, 5, 6.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Classifying numbers by their properties: Outcome Check

Outcome check for Classifying numbers by their properties: Explain the meaning of odd, even, square numbers, square root, prime numbers, multiples of numbers, factors of numbers.

1. List all factors of 18.
2. List the first five multiples of 6.
3. Circle any prime numbers in 2, 3, 4, 5, 6.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Fractions with the same denominator: Lesson Opener

Learning focus: Fractions with the same denominator

Country link: a Rwandan school, market, farming, building, transport, or data problem for Fractions with the same denominator.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- explain the meaning of fractions, add and subtract fractions with the same denominator, multiply and divide fractions accurately
- read, write, and recognise fractions, including the roles of the numerator and denominator in determining size
- explain and demonstrate how adding or subtracting fractions with the same denominator is like adding any other unit: metres, grams
- compare two fractions with the same denominator
- add and subtract fractions that have the same denominator
- solve and explain problems about fractions with the same denominator

Inquiry questions:

- How can fractions with the same denominator help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Fractions with the same denominator: Learn

Fractions with the same denominator teaches how parts of a whole are named and combined.

Core idea:

- The denominator tells how many equal parts make one whole.
- The numerator tells how many of those parts are being used.
- Fractions with the same denominator can be added or subtracted by working with the numerators.
- The denominator stays the same because the size of the parts has not changed.

Key words:

- Fractions with the same denominator
- Fractions
- with
- same
- denominator
- method
- model
- unit

Fractions with the same denominator: Worked Example

Worked example for Fractions with the same denominator: A learner eats $\frac{2}{8}$ of a pineapple in the morning and $\frac{3}{8}$ in the afternoon. What fraction is eaten?

1. The denominators are the same, so the parts are equal-sized eighths.
2. Add the numerators: $2 + 3 = 5$.
3. Keep the denominator: 8.

Answer: $\frac{5}{8}$ of the pineapple was eaten.

Fractions with the same denominator: Vocabulary And Study Skill

Use these words and study moves before you practise Fractions with the same denominator. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Fractions with the same denominator
- Fractions
- with
- same
- denominator
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Fractions with the same denominator without a clear example, method, or evidence.

Fractions with the same denominator: Guided Activity

Guided activity for Fractions with the same denominator:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Fractions with the same denominator: Practice And Correction

Practice for Fractions with the same denominator:

Main outcome: explain the meaning of fractions, add and subtract fractions with the same denominator, multiply and divide fractions accurately.

1. Draw one whole divided into 8 equal parts. Shade $\frac{3}{8}$.
2. Calculate $\frac{2}{7} + \frac{4}{7}$ and explain why the denominator stays 7.
3. Write one story problem using fractions with the same denominator, then solve it.

Answer check:

Answers: $\frac{2}{7} + \frac{4}{7} = \frac{6}{7}$. The denominator stays 7 because the equal parts are sevenths.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to fractions with the same denominator and solve it.

Fractions with the same denominator: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Fractions with the same denominator.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Fractions with the same denominator: Outcome Check

Outcome check for Fractions with the same denominator: Explain the meaning of fractions, add and subtract fractions with the same denominator, multiply and divide fractions accurately.

1. Draw $\frac{5}{8}$ on a fraction strip.
2. Calculate $\frac{3}{9} + \frac{4}{9}$.
3. Explain why the denominator does not change.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Fractions with the same denominator: Outcome Check

Outcome check for Fractions with the same denominator: Read, write, and recognise fractions, including the roles of the numerator and denominator in determining size.

1. Draw $\frac{5}{8}$ on a fraction strip.
2. Calculate $\frac{3}{9} + \frac{4}{9}$.
3. Explain why the denominator does not change.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Decimal numbers: Lesson Opener

Learning focus: Decimal numbers

Country link: a Rwandan school, market, farming, building, transport, or data problem for Decimal numbers.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- add, subtract and compare decimal numbers using place values of decimals up to 2 decimal places
- explain the concept of decimal numbers using place values up to 2 decimal places
- correctly read and write decimal numbers in figures and in words
- identify the place values in written decimals
- compare 2 or more decimal numbers using greater than, less than or equal ($>$, $<$ or $=$)
- accurately translate between written decimals and their spoken English forms

Inquiry questions:

- How can decimal numbers help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Decimal numbers: Learn

Decimal numbers teaches another way to write parts of a whole using place value.

Core idea:

- The first digit after the decimal point shows tenths.
- The second digit shows hundredths.
- Decimal numbers are compared from left to right, just like whole numbers.
- Money and measurements often use decimals.

Key words:

- Decimal numbers
- Decimal
- numbers
- method
- model
- unit
- answer
- check

Decimal numbers: Worked Example

Worked example for Decimal numbers: Compare 4.35 and 4.5.

1. Write 4.5 as 4.50 so both numbers show hundredths.
2. Compare whole numbers: both are 4.
3. Compare tenths: 3 tenths is less than 5 tenths.

Answer: 4.35 is less than 4.5.

Decimal numbers: Vocabulary And Study Skill

Use these words and study moves before you practise Decimal numbers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Decimal numbers
- Decimal
- numbers
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Decimal numbers without a clear example, method, or evidence.

Decimal numbers: Guided Activity

Guided activity for Decimal numbers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Decimal numbers: Practice And Correction

Practice for Decimal numbers:

Main outcome: add, subtract and compare decimal numbers using place values of decimals up to 2 decimal places.

1. Write 3 tenths as a decimal.
2. Order 2.05, 2.5, and 2.15 from smallest to largest.
3. Use a money or measurement example to compare two decimal numbers.

Answer check:

Answers: 3 tenths = 0.3. Order: 2.05, 2.15, 2.5.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to decimal numbers and solve it.

Decimal numbers: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Decimal numbers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Decimal numbers: Outcome Check

Outcome check for Decimal numbers: Add, subtract and compare decimal numbers using place values of decimals up to 2 decimal places.

1. Write 6 tenths as a decimal.
2. Compare 3.08 and 3.8 using place value.
3. Explain why your comparison is correct.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Decimal numbers: Outcome Check

Outcome check for Decimal numbers: Explain the concept of decimal numbers using place values up to 2 decimal places.

1. Write 6 tenths as a decimal.
2. Compare 3.08 and 3.8 using place value.
3. Explain why your comparison is correct.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Length measurements: Lesson Opener

Learning focus: Length measurements

Country link: a Rwandan school, market, farming, building, transport, or data problem for Length measurements.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- convert between units of length and apply this knowledge in solving mathematical problems related to daily life situations, including perimeters
- build on measurement of length
- state different units of length measurements
- explain the relationship between the units of length measurements
- explain the perimeter of a shape as the distance around it
- appropriately estimate length and provide a justification

Inquiry questions:

- How can length measurements help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Length measurements: Learn

Length measurements teaches how to measure distance or how long something is.

Core idea:

- Choose the unit that fits the object: millimetres, centimetres, metres, or kilometres.
- Convert by knowing the relationship between units.
- Estimate before measuring so you can notice unreasonable answers.

Key words:

- Length measurements
- Length
- measurements
- method
- model
- unit
- answer
- check

Length measurements: Worked Example

Worked example for Length measurements: A classroom is 8 m long and a corridor is 350 cm long. Which is longer?

1. Convert 8 m to centimetres: $8 \text{ m} = 800 \text{ cm}$.
2. Compare 800 cm and 350 cm.
3. 800 cm is greater than 350 cm.

Answer: The classroom is longer.

Length measurements: Vocabulary And Study Skill

Use these words and study moves before you practise Length measurements. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Length measurements
- Length
- measurements
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Length measurements without a clear example, method, or evidence.

Length measurements: Guided Activity

Guided activity for Length measurements:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Length measurements: Practice And Correction

Practice for Length measurements:

Main outcome: convert between units of length and apply this knowledge in solving mathematical problems related to daily life situations, including perimeters.

1. Estimate the length of a desk, then choose a suitable unit.
2. Convert 4 m to centimetres.
3. Solve: a path is 120 m and another is 75 m. What is the total length?

Answer check:

Answer: $4\text{ m} = 400\text{ cm}$. $120\text{ m} + 75\text{ m} = 195\text{ m}$.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to length measurements and solve it.

Length measurements: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Length measurements.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Length measurements: Outcome Check

Outcome check for Length measurements: Convert between units of length and apply this knowledge in solving mathematical problems related to daily life situations, including perimeters.

1. Convert 6 m to centimetres.
2. Find the total length of 45 m and 38 m.
3. State the unit clearly.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Length measurements: Outcome Check

Outcome check for Length measurements: Build on measurement of length.

1. Convert 6 m to centimetres.
2. Find the total length of 45 m and 38 m.
3. State the unit clearly.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Capacity Measurements: Lesson Opener

Learning focus: Capacity Measurements

Country link: a Rwandan school, market, farming, building, transport, or data problem for Capacity Measurements.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- convert between units of capacity and apply this knowledge in solving mathematical problems related to daily life situations
- distinguish between the capacities of different liquid containers through observation
- state the different units of capacity measurement
- explain the relationship between units of capacity measurement
- solve and explain problems about capacity measurements
- estimate the capacity of a liquid container through observation

Inquiry questions:

- How can capacity measurements help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Capacity Measurements: Learn

Capacity Measurements teaches how much liquid or material a container can hold.

Core idea:

- Capacity is measured with units such as millilitres and litres.
- Compare containers by reading the scale or using the same measuring container.
- Always state the unit with the number.

Key words:

- Capacity Measurements
- Capacity
- Measurements
- method
- model
- unit

Capacity Measurements: Worked Example

Worked example for Capacity Measurements: A jerrycan holds 20 litres. A bottle holds 2 litres. How many full bottles fill the jerrycan?

1. Use division because equal bottles fill one container.
2. $20 \text{ litres} / 2 \text{ litres} = 10$.
3. Check: $10 \text{ bottles} \times 2 \text{ litres} = 20 \text{ litres}$.

Answer: 10 bottles.

Capacity Measurements: Vocabulary And Study Skill

Use these words and study moves before you practise Capacity Measurements. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Capacity Measurements
- Capacity
- Measurements
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Capacity Measurements without a clear example, method, or evidence.

Capacity Measurements: Guided Activity

Guided activity for Capacity Measurements:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Capacity Measurements: Practice And Correction

Practice for Capacity Measurements:

Main outcome: convert between units of capacity and apply this knowledge in solving mathematical problems related to daily life situations.

1. Name two containers measured in litres.
2. Convert 3 litres to millilitres.
3. A bucket holds 15 litres. How many 5-litre containers fill it?

Answer check:

Answer: 3 litres = 3000 millilitres. $15 \text{ litres} / 5 \text{ litres} = 3 \text{ containers}$.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to capacity measurements and solve it.

Capacity Measurements: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Capacity Measurements.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Capacity Measurements: Outcome Check

Outcome check for Capacity Measurements: Convert between units of capacity and apply this knowledge in solving mathematical problems related to daily life situations.

1. Convert 5 litres to millilitres.
2. Solve: a 12-litre container is filled by 3-litre bottles. How many bottles?
3. Check the answer by multiplication.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Capacity Measurements: Outcome Check

Outcome check for Capacity Measurements: Distinguish between the capacities of different liquid containers through observation.

1. Convert 5 litres to millilitres.
2. Solve: a 12-litre container is filled by 3-litre bottles. How many bottles?
3. Check the answer by multiplication.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Mass measurements: Lesson Opener

Learning focus: Mass measurements

Country link: a Rwandan school, market, farming, building, transport, or data problem for Mass measurements.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- convert between units of mass and apply this knowledge in solving mathematical problems related to daily life situations
- identify different units of mass and relate them to appropriate objects, recognizing how these units are connected
- estimate the masses of objects through observation and explain the process of solving mathematical problems involving mass measurement
- select and correctly use appropriate scales and materials to measure mass in daily situations
- accurately read, write, and convert mass units, and use addition, subtraction, and multiplication to solve related problems
- solve and explain problems about mass measurements

Inquiry questions:

- How can mass measurements help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Mass measurements: Learn

Mass measurements teaches how heavy objects are.

Core idea:

- Mass is measured with units such as grams and kilograms.
- A balance or scale helps compare mass fairly.
- Convert units before adding or comparing if the units are different.

Key words:

- Mass measurements
- Mass
- measurements
- method
- model
- unit
- answer
- check

Mass measurements: Worked Example

Worked example for Mass measurements: A bag of beans has a mass of 3 kg. A small packet has a mass of 500 g. What is the total mass in grams?

1. Convert 3 kg to grams: $3 \text{ kg} = 3000 \text{ g}$.
2. Add $3000 \text{ g} + 500 \text{ g} = 3500 \text{ g}$.

Answer: 3500 g.

Mass measurements: Vocabulary And Study Skill

Use these words and study moves before you practise Mass measurements. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Mass measurements
- Mass
- measurements
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Mass measurements without a clear example, method, or evidence.

Mass measurements: Guided Activity

Guided activity for Mass measurements:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Mass measurements: Practice And Correction

Practice for Mass measurements:

Main outcome: convert between units of mass and apply this knowledge in solving mathematical problems related to daily life situations.

1. Name two objects measured in grams and two measured in kilograms.
2. Convert 2 kg to grams.
3. Compare 1500 g and 2 kg. Which is heavier?

Answer check:

Answer: $2\text{ kg} = 2000\text{ g}$, so 2 kg is heavier than 1500 g.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to mass measurements and solve it.

Mass measurements: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Mass measurements.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Mass measurements: Outcome Check

Outcome check for Mass measurements: Convert between units of mass and apply this knowledge in solving mathematical problems related to daily life situations.

1. Convert 4 kg to grams.
2. Compare 2500 g and 3 kg.
3. Write the heavier mass with a reason.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Mass measurements: Outcome Check

Outcome check for Mass measurements: Identify different units of mass and relate them to appropriate objects, recognizing how these units are connected.

1. Convert 4 kg to grams.
2. Compare 2500 g and 3 kg.
3. Write the heavier mass with a reason.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Area and land measurements: Lesson Opener

Learning focus: Area and land measurements

Country link: a Rwandan school, market, farming, building, transport, or data problem for Area and land measurements.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- determine area of shapes and use square and land units in solving related problems
- explain the concept of square units
- state the units of area and land measurements
- establish the relationship between the units of area measurements
- list the units of area and land measurements in ascending and descending order
- practically differentiate the measurement of area from the measurement of length

Inquiry questions:

- How can area and land measurements help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Area and land measurements: Learn

Area and land measurements teaches how much flat surface is covered.

Core idea:

- Area is counted in square units.
- Rectangles can be found by length x width.
- Land and floor measurements need clear units and a labelled diagram.

Key words:

- Area and land measurements
- Area
- land
- measurements
- method
- model
- unit
- answer

Area and land measurements: Worked Example

Worked example for Area and land measurements: A rectangular garden is 9 m long and 4 m wide. Find its area.

1. Draw a rectangle and label length 9 m and width 4 m.
2. Area of a rectangle = length x width.
3. $9 \times 4 = 36$.

Answer: 36 square metres.

Area and land measurements: Vocabulary And Study Skill

Use these words and study moves before you practise Area and land measurements. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Area and land measurements
- Area
- land
- measurements
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Area and land measurements without a clear example, method, or evidence.

Area and land measurements: Guided Activity

Guided activity for Area and land measurements:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Area and land measurements: Practice And Correction

Practice for Area and land measurements:

Main outcome: determine area of shapes and use square and land units in solving related problems.

1. Draw a rectangle that is 6 units long and 3 units wide.
2. Find its area in square units.
3. Explain why area uses square units.

Answer check:

Answer: a 6 by 3 rectangle has area 18 square units.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to area and land measurements and solve it.

Area and land measurements: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Area and land measurements.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Area and land measurements: Outcome Check

Outcome check for Area and land measurements: Determine area of shapes and use square and land units in solving related problems.

1. Find the area of a rectangle 7 units by 5 units.
2. Explain why the answer uses square units.
3. Draw a quick labelled sketch.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Area and land measurements: Outcome Check

Outcome check for Area and land measurements: Explain the concept of square units.

1. Find the area of a rectangle 7 units by 5 units.
2. Explain why the answer uses square units.
3. Draw a quick labelled sketch.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Time: Lesson Opener

Learning focus: Time

Country link: a Rwandan school, market, farming, building, transport, or data problem for Time.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- tell, write and convert time appropriately
- recognise different units of time
- explain the process of solving mathematical problems involving time
- explain the meaning of am and pm
- read and tell the time accurately
- convert between units of time

Inquiry questions:

- How can time help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Time: Learn

Time teaches how to read, write, convert, and use time.

Core idea:

- Time can be shown on analogue clocks, digital clocks, calendars, and timetables.
- 60 seconds make 1 minute, and 60 minutes make 1 hour.
- Timetables help you find start time, end time, and duration.

Key words:

- Time
- method
- model
- unit
- answer
- check

Time: Worked Example

Worked example for Time: A lesson starts at 8:10 and ends at 8:50. How long is the lesson?

1. Count from 8:10 to 8:50.
2. 10 minutes to 8:20, 20 more to 8:40, 10 more to 8:50.
3. Total time = 40 minutes.

Answer: 40 minutes.

Time: Vocabulary And Study Skill

Use these words and study moves before you practise Time. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Time
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Time without a clear example, method, or evidence.

Time: Guided Activity

Guided activity for Time:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Time: Practice And Correction

Practice for Time:

Main outcome: tell, write and convert time appropriately.

1. Read a clock showing 7:30 and write the time in words.
2. Convert 2 hours to minutes.
3. A trip starts at 9:15 and ends at 10:00. Find the duration.

Answer check:

Answers: 2 hours = 120 minutes. 9:15 to 10:00 is 45 minutes.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to time and solve it.

Time: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Time.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Time: Outcome Check

Outcome check for Time: Tell, write and convert time appropriately.

1. Convert 3 hours to minutes.
2. Find the duration from 8:20 to 9:05.
3. Write the answer in minutes.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Time: Outcome Check

Outcome check for Time: Recognise different units of time.

1. Convert 3 hours to minutes.
2. Find the duration from 8:20 to 9:05.
3. Write the answer in minutes.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Number patterns: Lesson Opener

Learning focus: Number patterns

Country link: a Rwandan school, market, farming, building, transport, or data problem for Number patterns.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- describe and generate number patterns following a rule
- explain how to arrange whole numbers according to their size in increasing and decreasing order
- explain the meaning of arithmetic and geometric progression
- determine the clue or pattern for a given arithmetic progression/ geometric progression
- differentiate between arithmetic progression and geometric progression
- arrange whole numbers in different orders

Inquiry questions:

- How can number patterns help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Number patterns: Learn

Number patterns teaches how numbers change in a regular way.

Core idea:

- Look at how each term changes to the next term.
- A pattern can add, subtract, multiply, divide, or combine steps.
- Write the rule before extending the pattern.

Key words:

- Number patterns
- Number
- patterns
- method
- model

Number patterns: Worked Example

Worked example for Number patterns: Continue the pattern 6, 12, 18, 24, __, __.

1. Compare terms: $12 - 6 = 6$, $18 - 12 = 6$.
2. Rule: add 6 each time.
3. $24 + 6 = 30$, $30 + 6 = 36$.

Answer: 30, 36.

Number patterns: Vocabulary And Study Skill

Use these words and study moves before you practise Number patterns. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Number patterns
- Number
- patterns
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Number patterns without a clear example, method, or evidence.

Number patterns: Guided Activity

Guided activity for Number patterns:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Number patterns: Practice And Correction

Practice for Number patterns:

Main outcome: describe and generate number patterns following a rule.

1. Continue the pattern 5, 10, 15, 20, __, __.
2. Write the rule for the pattern.
3. Create a similar pattern that increases by 4 each time.

Answer check:

Answers: 25, 30. Rule: add 5 each time.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to number patterns and solve it.

Number patterns: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Number patterns.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Number patterns: Outcome Check

Outcome check for Number patterns: Describe and generate number patterns following a rule.

1. Continue 7, 14, 21, 28, __, __.
2. Write the rule.
3. Create one new pattern with the same rule.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Number patterns: Outcome Check

Outcome check for Number patterns: Explain how to arrange whole numbers according to their size in increasing and decreasing order.

1. Continue 7, 14, 21, 28, __, __.
2. Write the rule.
3. Create one new pattern with the same rule.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving missing number in addition and subtraction: Lesson Opener

Learning focus: Solving problems involving missing number in addition and subtraction

Country link: a Rwandan school, market, farming, building, transport, or data problem for Solving problems involving missing number in addition and subtraction.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve missing number problems involving addition and subtraction
- use relationships between numbers to solve missing number problems
- solve and explain problems about solving problems involving missing number in addition and subtraction
- solving simple missing number problems involving addition
- solving simple missing number problems involving subtraction
- solving simple missing number problems involving both addition and subtraction

Inquiry questions:

- How can solving problems involving missing number in addition and subtraction help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving missing number in addition and subtraction: Learn

Solving problems involving missing number in addition and subtraction teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Solving problems involving missing number in addition and subtraction
- Solving
- problems
- involving
- missing
- number
- addition
- subtraction

Solving problems involving missing number in addition and subtraction: Worked Example

Worked example for Solving problems involving missing number in addition and subtraction: Find the missing number: $45 + \underline{\quad} = 73$.

1. Use subtraction to undo addition.
2. $73 - 45 = 28$.
3. Check: $45 + 28 = 73$.

Answer: 28.

Solving problems involving missing number in addition and subtraction: Vocabulary And Study Skill

Use these words and study moves before you practise Solving problems involving missing number in addition and subtraction. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Solving problems involving missing number in addition and subtraction
- Solving
- problems
- involving
- missing
- number
- addition
- subtraction

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Solving problems involving missing number in addition and subtraction without a clear example, method, or evidence.

Solving problems involving missing number in addition and subtraction: Guided Activity

Guided activity for Solving problems involving missing number in addition and subtraction:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Solving problems involving missing number in addition and subtraction: Practice And Correction

Practice for Solving problems involving missing number in addition and subtraction:

Main outcome: solve missing number problems involving addition and subtraction.

1. Find the missing number: $36 + \underline{\quad} = 81$.
2. Find the missing number: $\underline{\quad} - 18 = 47$.
3. Check each answer using the inverse operation.

Answer check:

Answers: 45 and 65. Checks: $36 + 45 = 81$; $65 - 18 = 47$.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to solving problems involving missing number in addition and subtraction and solve it.

Solving problems involving missing number in addition and subtraction: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Solving problems involving missing number in addition and subtraction.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving missing number in addition and subtraction: Outcome Check

Outcome check for Solving problems involving missing number in addition and subtraction: Solve missing number problems involving addition and subtraction.

1. Find $52 + \underline{\quad} = 90$.
2. Find $\underline{\quad} - 25 = 46$.
3. Check both answers with inverse operations.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving missing number in addition and subtraction: Outcome Check

Outcome check for Solving problems involving missing number in addition and subtraction: Use relationships between numbers to solve missing number problems.

1. Find $52 + \underline{\quad} = 90$.
2. Find $\underline{\quad} - 25 = 46$.
3. Check both answers with inverse operations.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Lines and angles: Lesson Opener

Learning focus: Lines and angles

Country link: a Rwandan school, market, farming, building, transport, or data problem for Lines and angles.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- identify types of lines and angles and use a protractor to measure angles
- recognize different types of lines and angles, identifying their properties and distinctions
- select appropriate tools to measure and classify angles accurately in geometric tasks
- practice drawing straight lines and measuring line segments accurately using a ruler
- correctly measure different angles using a protractor and apply appropriate instruments for geometric tasks
- differentiate types of angles based on their properties and develop skills in analyzing geometric shapes

Inquiry questions:

- How can lines and angles help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Lines and angles: Learn

Lines and angles teaches how lines and angles describe shapes and directions.

Core idea:

- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:

- Lines and angles
- Lines
- angles
- method
- model
- unit
- answer
- check

Lines and angles: Worked Example

Worked example for Lines and angles: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Lines and angles: Vocabulary And Study Skill

Use these words and study moves before you practise Lines and angles. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Lines and angles
- Lines
- angles
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Lines and angles without a clear example, method, or evidence.

Lines and angles: Guided Activity

Guided activity for Lines and angles:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Lines and angles: Practice And Correction

Practice for Lines and angles:

Main outcome: identify types of lines and angles and use a protractor to measure angles.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:

Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to lines and angles and solve it.

Lines and angles: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Lines and angles.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Lines and angles: Outcome Check

Outcome check for Lines and angles: Identify types of lines and angles and use a protractor to measure angles.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Lines and angles: Outcome Check

Outcome check for Lines and angles: Recognize different types of lines and angles, identifying their properties and distinctions.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Shapes in the plane: Lesson Opener

Learning focus: Shapes in the plane

Country link: a Rwandan school, market, farming, building, transport, or data problem for Shapes in the plane.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- use geometric properties, including symmetry, to sort shapes
- name triangles and special quadrilaterals
- state the geometric properties that can be used to distinguish between shapes
- identify triangles and special quadrilaterals in everyday situations
- select appropriate geometric instruments to construct triangles, quadrilaterals and examine their properties
- distinguish between different shapes using their properties

Inquiry questions:

- How can shapes in the plane help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Shapes in the plane: Learn

Shapes in the plane teaches properties of two-dimensional shapes.

Core idea:

- Shapes can be classified by sides, corners, lines of symmetry, and angles.
- A labelled sketch helps you compare shapes accurately.
- Use correct names such as triangle, rectangle, square, and circle.

Key words:

- Shapes in the plane
- Shapes
- plane
- method
- model
- unit
- answer
- check

Shapes in the plane: Worked Example

Worked example for Shapes in the plane: Classify a shape with 4 equal sides and 4 right angles.

1. Four sides means it is a quadrilateral.
2. Equal sides and right angles match a square.
3. Check the name and properties.

Answer: The shape is a square.

Shapes in the plane: Vocabulary And Study Skill

Use these words and study moves before you practise Shapes in the plane. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Shapes in the plane
- Shapes
- plane
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Shapes in the plane without a clear example, method, or evidence.

Shapes in the plane: Guided Activity

Guided activity for Shapes in the plane:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Shapes in the plane: Practice And Correction

Practice for Shapes in the plane:

Main outcome: use geometric properties, including symmetry, to sort shapes.

1. Draw a triangle, rectangle, and square.
2. Write two properties for each shape.
3. Circle the shape that has four equal sides and four right angles.

Answer check:

Answer: the square has four equal sides and four right angles.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to shapes in the plane and solve it.

Shapes in the plane: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Shapes in the plane.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Shapes in the plane: Outcome Check

Outcome check for Shapes in the plane: Use geometric properties, including symmetry, to sort shapes.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Shapes in the plane: Outcome Check

Outcome check for Shapes in the plane: Name triangles and special quadrilaterals.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Area of shapes in the plane: Lesson Opener

Learning focus: Area of shapes in the plane

Country link: a Rwandan school, market, farming, building, transport, or data problem for Area of shapes in the plane.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- use rectangles to determine the area of triangles and special quadrilaterals
- explain area as the space enclosed within a perimeter
- distinguish between area and perimeter
- select the correct units to use
- practically determine the area of triangles and quadrilaterals by relating them to the area of rectangles
- solve problems involving the determination of the area of various 2D shapes

Inquiry questions:

- How can area of shapes in the plane help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Area of shapes in the plane: Learn

Area of shapes in the plane teaches how much flat surface is covered.

Core idea:

- Area is counted in square units.
- Rectangles can be found by length x width.
- Land and floor measurements need clear units and a labelled diagram.

Key words:

- Area of shapes in the plane
- Area
- shapes
- plane
- method
- model
- unit
- answer

Area of shapes in the plane: Worked Example

Worked example for Area of shapes in the plane: A rectangular garden is 9 m long and 4 m wide. Find its area.

1. Draw a rectangle and label length 9 m and width 4 m.
2. Area of a rectangle = length x width.
3. $9 \times 4 = 36$.

Answer: 36 square metres.

Area of shapes in the plane: Vocabulary And Study Skill

Use these words and study moves before you practise Area of shapes in the plane. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Area of shapes in the plane
- Area
- shapes
- plane
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Area of shapes in the plane without a clear example, method, or evidence.

Area of shapes in the plane: Guided Activity

Guided activity for Area of shapes in the plane:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Area of shapes in the plane: Practice And Correction

Practice for Area of shapes in the plane:

Main outcome: use rectangles to determine the area of triangles and special quadrilaterals.

1. Draw a rectangle that is 6 units long and 3 units wide.
2. Find its area in square units.
3. Explain why area uses square units.

Answer check:

Answer: a 6 by 3 rectangle has area 18 square units.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to area of shapes in the plane and solve it.

Area of shapes in the plane: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Area of shapes in the plane.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Area of shapes in the plane: Outcome Check

Outcome check for Area of shapes in the plane: Use rectangles to determine the area of triangles and special quadrilaterals.

1. Find the area of a rectangle 7 units by 5 units.
2. Explain why the answer uses square units.
3. Draw a quick labelled sketch.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Area of shapes in the plane: Outcome Check

Outcome check for Area of shapes in the plane: Explain area as the space enclosed within a perimeter.

1. Find the area of a rectangle 7 units by 5 units.
2. Explain why the answer uses square units.
3. Draw a quick labelled sketch.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Mathematical operations on whole numbers up to 100,000: Review Clinic 1

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Mathematical operations on whole numbers up to 100,000 that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Positive and negative integers: Review Clinic 2

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Positive and negative integers that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Classifying numbers by their properties: Review Clinic 3

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Classifying numbers by their properties that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Fractions with the same denominator: Review Clinic 4

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Fractions with the same denominator that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Mathematics.

- Mathematical operations on whole numbers up to 100,000: Explain this term using an example from the book, your school, or your community.
- Positive and negative integers: Explain this term using an example from the book, your school, or your community.
- Classifying numbers by their properties: Explain this term using an example from the book, your school, or your community.
- Fractions with the same denominator: Explain this term using an example from the book, your school, or your community.
- Decimal numbers: Explain this term using an example from the book, your school, or your community.
- Length measurements: Explain this term using an example from the book, your school, or your community.
- Capacity Measurements: Explain this term using an example from the book, your school, or your community.
- Mass measurements: Explain this term using an example from the book, your school, or your community.
- Area and land measurements: Explain this term using an example from the book, your school, or your community.
- Time: Explain this term using an example from the book, your school, or your community.
- Number patterns: Explain this term using an example from the book, your school, or your community.
- Solving problems involving missing number in addition and subtraction: Explain this term using an example from the book, your school, or your community.
- Lines and angles: Explain this term using an example from the book, your school, or your community.
- Shapes in the plane: Explain this term using an example from the book, your school, or your community.
- Area of shapes in the plane: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Mathematics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.